

BarkSense.

An on-device dog-vocalization classifier on a XIAO ESP32S3. A custom DS-CNN — trained from scratch in Keras, quantized with `tf.lite`, run on-chip via TFLite Micro — labels 1-second audio windows as **bark · growl · grunt · ambient** and acts locally.

No audio recorded or transmitted — only a class label + timestamp ever leave the device. No cloud, no Edge Impulse, no pre-trained models.



Design Targets — scored honestly

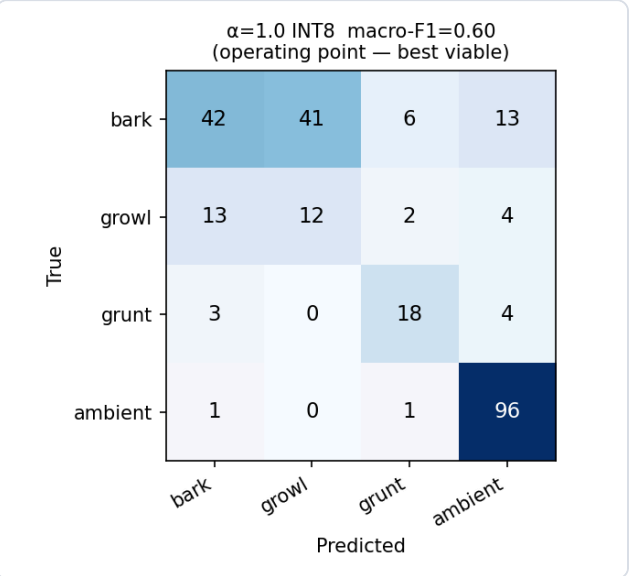
METRIC	TARGET	ACTUAL
Macro-F1 (4-class)	≥ 85%	60% ✗
Bark recall	≥ 82%	41% ✗
Latency / window	≤ 500 ms	~150 ms* ~
Battery life	≥ 12 h	untested ·
Flash footprint	≤ 300 KB	64 KB ✓
Peak RAM	≤ 100 KB	1.8 MB ✗

*feature extraction; α=1.0 `Invoke` intermittently stalls — see journey. Values for α=1.0 INT8 (best viable model).

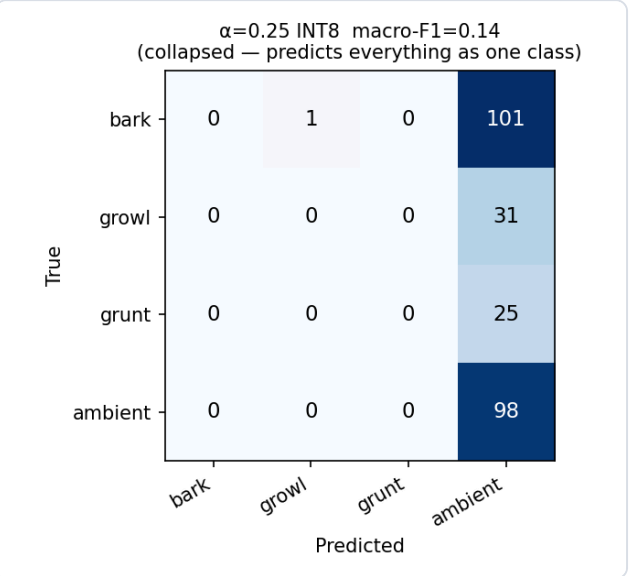
The two compute targets split: **flash passes easily**, but the only accurate-enough model needs **18× the RAM budget**, forcing it into PSRAM. That single fact drives the whole project.

Operating Point

There is no operating point that meets every target. Best viable accuracy = α=1.0 INT8 (F1 0.60); it stalls on hardware, so the **demo ships the small α=0.25** (stable, real-time, accepts lower accuracy).

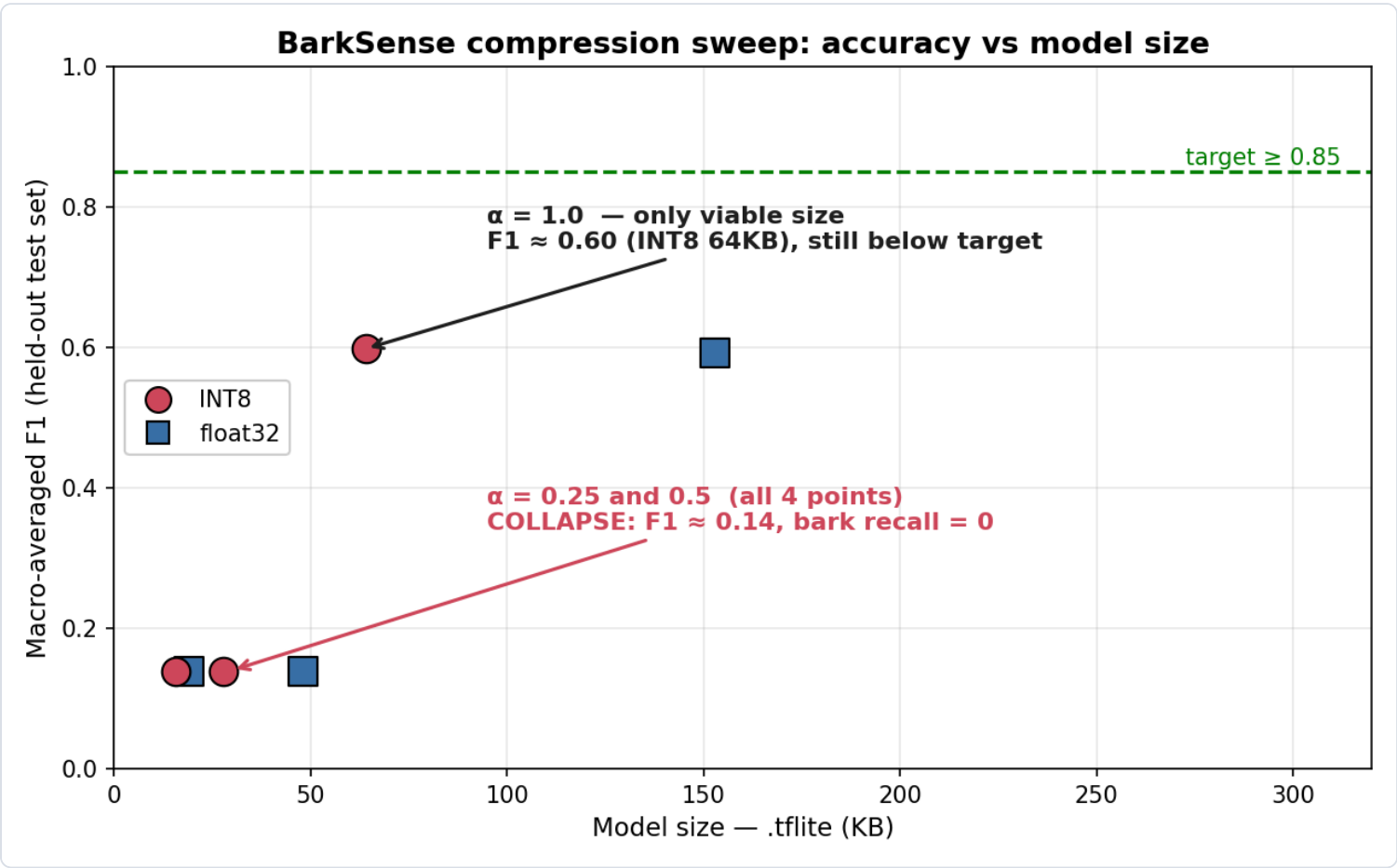


α=1.0 INT8 — ambient strong (0.98); main error is bark↔growl.



α=0.25 — collapse: everything → one class.

Trade-off Sweep — core deliverable



DS-CNN width α ∈ {0.25, 0.5, 1.0} × {float32, INT8}, on the locked held-out test set.

- α = 0.25 and 0.5 collapse — macro-F1 ≈ 0.14, **bark recall = 0**. Too little capacity for this 4-class task on the available data.
- Only α = 1.0 is viable (INT8: F1 ≈ 0.60, 64 KB) — still below the 85% target.
- INT8 is free here: F1 0.592→0.598 while size 153→64 KB and RAM 7.3→1.8 MB.

The viable-accuracy region and the RAM-budget region **do not overlap** — there is no point on this curve that satisfies both.

Diagnostic Journey

FAILURE	ROOT CAUSE	UNLOCK
Small models are dead — predict one class, bark recall 0	4-class task exceeds tiny-model capacity	Move to α=1.0 (F1 0.60) — breaks RAM budget
On device, <i>everything</i> reads "dog 0.97" — constant false alarms	PDM mic ~1300 DC offset; training audio is DC-free → power_to_db (ref=max) swamped → degenerate features	Subtract DC per clip → real bark scores 0.70, ambient stays ambient
α=1.0 inference freezes for minutes	1.8 MB arena in OPI PSRAM → Invoke () intermittently stalls	Deploy small α=0.25 (0.46 MB) → stable real-time; trade accuracy for a working demo

Hardware

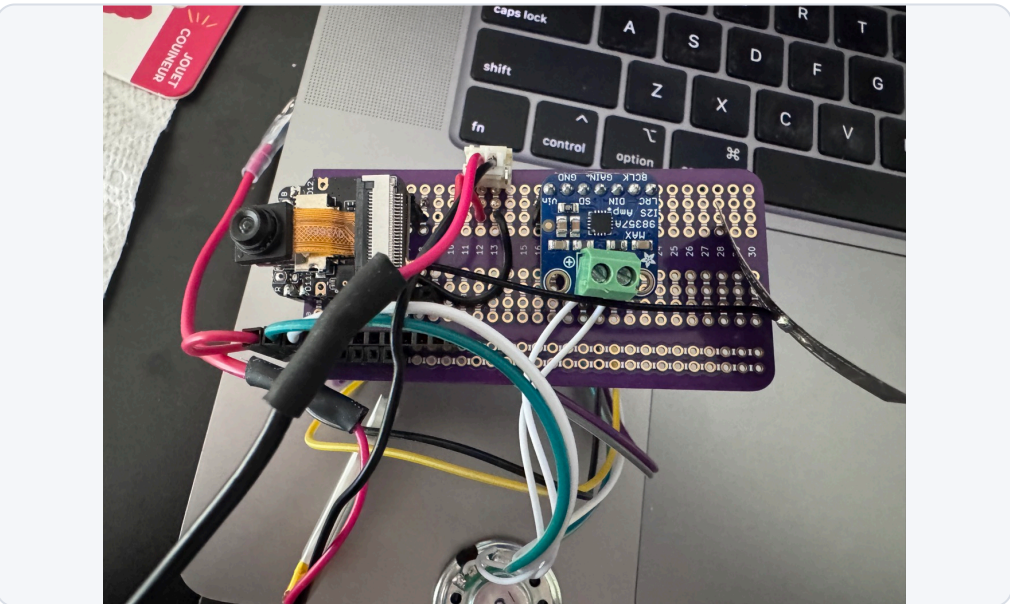
Board	Seeed XIAO ESP32S3 Sense — 240 MHz, 8 MB flash, 8 MB OPI PSRAM
Mic	onboard PDM (I2S0): CLK 42, DATA 41
Speaker	MAX98357A (I2S1): BCLK D0, LRCLK D1, DIN D2
LED / Power	GPIO 21 · 500 mAh LiPo · 4 MB LittleFS for owner voice

Status & Next Steps

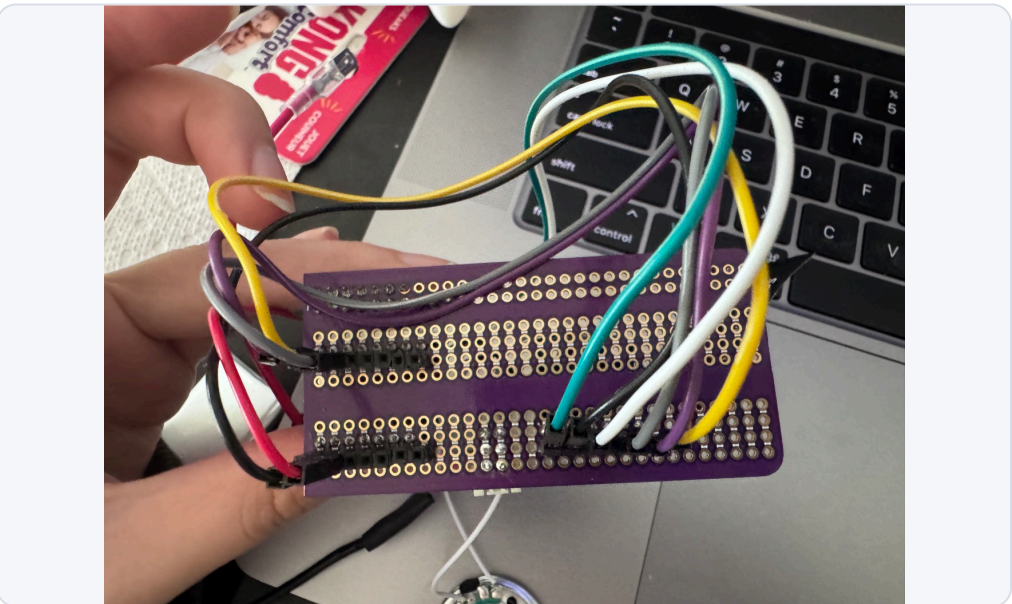
Works today: full on-device pipeline; live bark/growl → LED + 1.5 s owner-voice + phone push; quiet ignored; real-time, no stalls (α=0.25).

- PSRAM/inference stability for α=1.0 — the real blocker.
- Train a **middle model (α≈0.75)** — possibly stable *and* accurate.
- More data → lift bark recall, fix bark↔growl; re-add human-as-ambient.
- Formal latency + battery measurement.

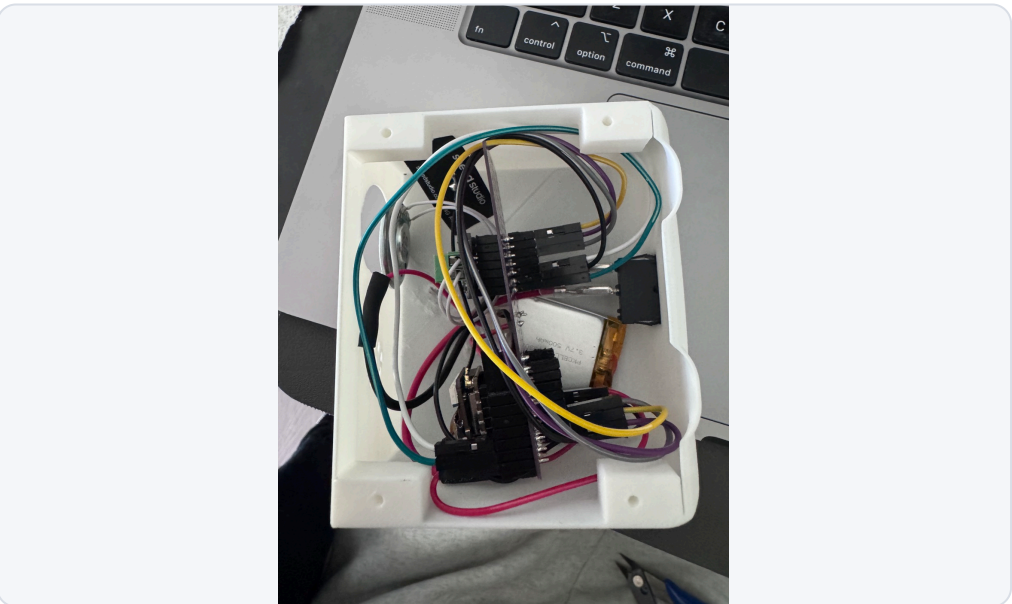
Hardware & Enclosure



Assembled board — front



Board — wiring (back)



Inside the enclosure



Finished enclosure